

CLAIMS

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TITLE: ZERO-TRUST DATA SANITIZATION (ZTDS) ARCHITECTURE FOR EPHEMERAL IN-MEMORY PII REDACTION AND CRYPTOGRAPHIC ENCLAVE ATTESTATION IN ARTIFICIAL INTELLIGENCE WORKFLOWS

WHAT IS CLAIMED IS:

1. A computer-implemented method for zero-trust data sanitization in artificial intelligence workflows, comprising: intercepting, by an endpoint runtime executing in volatile memory of a client computing device, an input query comprising unstructured text prior to serialization onto an external network; identifying, using one or more pattern recognition automata, one or more sensitive entities within the input query; substituting each identified sensitive entity with a unique surrogate token to generate a sanitized query, while maintaining an association between each surrogate token and its corresponding sensitive entity exclusively within an ephemeral, in-memory session map in volatile memory; validating cryptographic hardware attestation evidence received from a remote hardware-isolated execution enclave, the attestation evidence comprising measurements of software executing within the hardware-isolated execution enclave; transmitting, upon successful validation of the attestation evidence under a fail-closed policy, the sanitized query to the hardware-isolated execution enclave; receiving, from the hardware-isolated execution enclave, a tokenized output response; and rehydrating the tokenized output response at the client computing device by replacing each surrogate token with its corresponding sensitive entity retrieved from the ephemeral, in-memory session map.
2. The method of claim 1, wherein substituting each identified sensitive entity is performed in descending order of starting character offsets within the input query, thereby preventing character offset distortion during string replacement.
3. The method of claim 1, wherein the ephemeral, in-memory session map is strictly isolated to a single browser execution context or application process and is irrevocably destroyed upon closure of the execution context without persisting sensitive entity data to non-volatile storage.
4. The method of claim 1, wherein the cryptographic hardware attestation evidence comprises a cryptographically signed document generated by a hardware security module, the document including a fresh challenge nonce and platform configuration registers (PCR0, PCR1, PCR2) reflecting a cryptographic digest of an enclave image format (EIF).
5. The method of claim 4, wherein validating the attestation evidence comprises confirming that the PCR0 measurement matches a predetermined authorized hash value and that the challenge nonce matches an unexpired challenge issued by a policy decision point.

6. The method of claim 1, wherein communication with the hardware-isolated execution enclave is conducted over a local virtual socket channel (vsock) bypassing external network interfaces.
7. The method of claim 1, wherein the pattern recognition automata comprise a plurality of pre-compiled deterministic finite automata (DFA) corresponding to personally identifiable information (PII), protected health information (PHI), financial account credentials, and software secrets.
8. A secure endpoint data sanitization system, comprising: one or more processors; and a volatile memory coupled to the one or more processors, the volatile memory storing instructions that, when executed by the one or more processors, cause the system to: detect, in unstructured application text, sensitive data entities using compiled automata matchers; replace the detected sensitive data entities with unique surrogate tokens to construct a sanitized application payload; record mapping associations between the surrogate tokens and the detected sensitive data entities strictly within an ephemeral in-memory dictionary; dispatch the sanitized application payload to an attested confidential hardware enclave only upon verification of enclave attestation evidence against an authorized policy; and rehydrate surrogate tokens in an enclave response using the ephemeral in-memory dictionary.
9. A non-transitory computer-readable storage medium comprising instructions that, when executed by a client computing system, cause the system to perform the method of claim 1.
10. The method of claim 1, further comprising: serializing the ephemeral, in-memory session map at the client computing device; deriving a symmetric encryption key from an enterprise passphrase using a memory-hard password hashing function comprising Argon2id; encrypting the serialized session map using an authenticated encryption with associated data (AEAD) cipher comprising XChaCha20-Poly1305 with an ephemeral random nonce and salt, to construct an authenticated encrypted handoff payload; and transmitting the encrypted handoff payload to a second client computing device via a client-side uniform resource identifier (URI) fragment or encrypted file container, wherein the encrypted handoff payload bypasses intermediate server persistence and database storage.
11. The method of claim 10, further comprising: receiving the encrypted handoff payload at the second client computing device; deriving the symmetric encryption key at the second client computing device using the memory-hard password hashing function; decrypting the encrypted handoff payload into an ephemeral, volatile memory session map of the second client computing device; erasing the derived symmetric encryption key from memory of the second client computing device; and rehydrating tokenized artificial intelligence responses at the second client computing device by replacing surrogate tokens with sensitive entities retrieved from the decrypted ephemeral session map.
12. A zero-server collaborative data sanitization system, comprising: a first client computing device comprising a first processor and a first volatile memory, configured to: sanitize an artificial intelligence query by substituting sensitive entities with surrogate tokens; store associations between the sensitive entities and surrogate tokens in a first volatile in-memory dictionary; derive a symmetric key using Argon2id; encrypt the first in-memory dictionary using XChaCha20-Poly1305 authenticated

encryption into an encrypted handoff container; and export the encrypted handoff container via a client-side URI fragment; and a second client computing device comprising a second processor and a second volatile memory, configured to: import the encrypted handoff container from the URI fragment; decrypt the container into a second volatile in-memory dictionary in the second volatile memory; cryptographically wipe the symmetric key; and rehydrate tokenized responses from an artificial intelligence model using the second volatile in-memory dictionary.